

0.1 Pseudopotential Kohn–Sham discretization convergence

We now verify the accuracy and performance of SPARC-atomSFE for pseudopotential Kohn–Sham DFT calculations. Unless otherwise specified, we use an exponential mesh with shift $c = 0$, concentration parameter $a = 20$, domain size $R_{max} = 40$ Bohr, $N_{fe} = 10$ finite elements, polynomial degree $p = 20$, and quadrature order $q = 60$. We employ the SPMS suite of ONCV pseudopotentials, which include nonlinear core corrections.

We begin by studying convergence with respect to the domain size R_{max} and the number of finite elements N_{fe} . In particular, Figure 1 shows the convergence of the total energy and occupied valence eigenvalues for the PBE functional across atoms with atomic numbers $Z = 1\text{--}57$ and $Z = 72\text{--}83$, with reference results computed using $R_{max} = 40$ Bohr and $N_{fe} = 16$. We observe that both quantities converge exponentially with increasing R_{max} , with errors below 10^{-6} Ha achieved at $R_{max} \sim 27$ Bohr. We also observe rapid convergence with increasing N_{fe} , with accuracies better than 10^{-6} Ha attained at $N_{fe} \sim 9$. These results are consistent with previous GGA results obtained using a spectral scheme based on Chebyshev polynomials. We also find similar convergence behavior for all other exchange–correlation functionals considered here, with the exception of RPA-OEP, which has not been tested in the pseudopotential context.

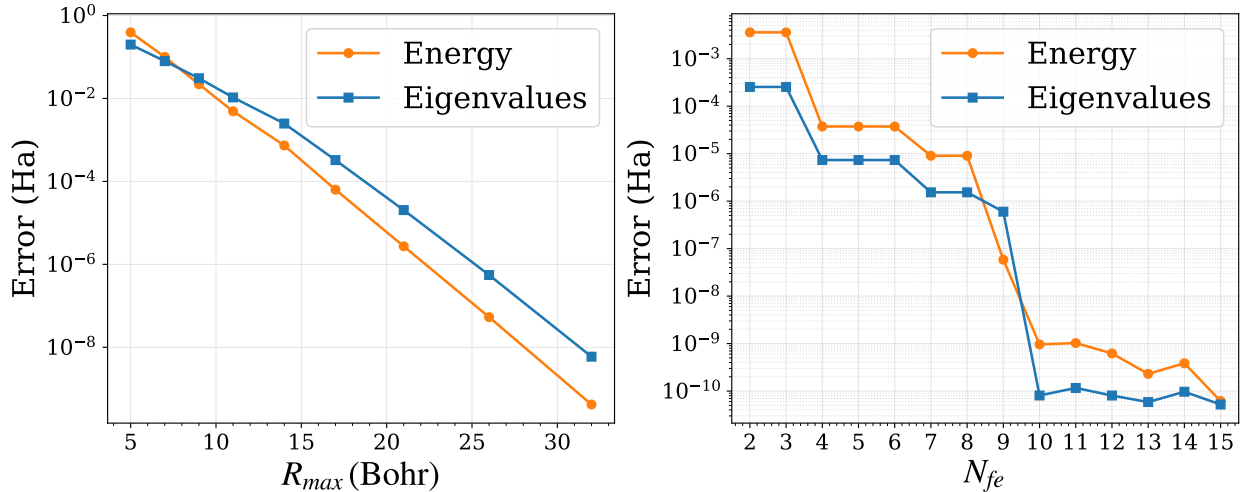


Figure 1: Convergence of the total energy and occupied valence eigenvalues with respect to domain size R_{max} (left) and number of finite elements N_{fe} (right) in pseudopotential Kohn–Sham DFT calculations, using the PBE exchange–correlation functional, for atomic numbers $Z = 1\text{--}57$ and $Z = 72\text{--}83$. The energy error is defined as the maximum absolute total-energy difference over all atoms in the set, while the eigenvalue error is defined as the maximum, over those atoms, of the per-atom mean absolute difference in the occupied valence eigenvalues. Reference values are computed using domain size $R_{max} = 40$ Bohr and $N_{fe} = 16$ finite elements.

As supplementary material to the convergence discussion presented in the main paper, the pseudopotential sweeps likewise use *nested* subsets of a common reference exponential mesh, rather than regenerating independent meshes for each choice of R_{max} or N_{fe} . This construction isolates the effect of radial resolution while preserving the underlying exponential mesh distribution across discretization levels. Figure 2 illustrates this subset construction for the pseudopotential case.

Finally, Figures 3, 4, and 5 give LDA-SVWN, rSCAN, and PBE0 counterparts to Figure 1: the same two-panel layout, error definitions, reference mesh ($R_{max} = 40$ Bohr, $N_{fe} = 16$), and atom ranges $Z = 1\text{--}57$ and $Z = 72\text{--}83$, with discretization trends consistent with the PBE panels in

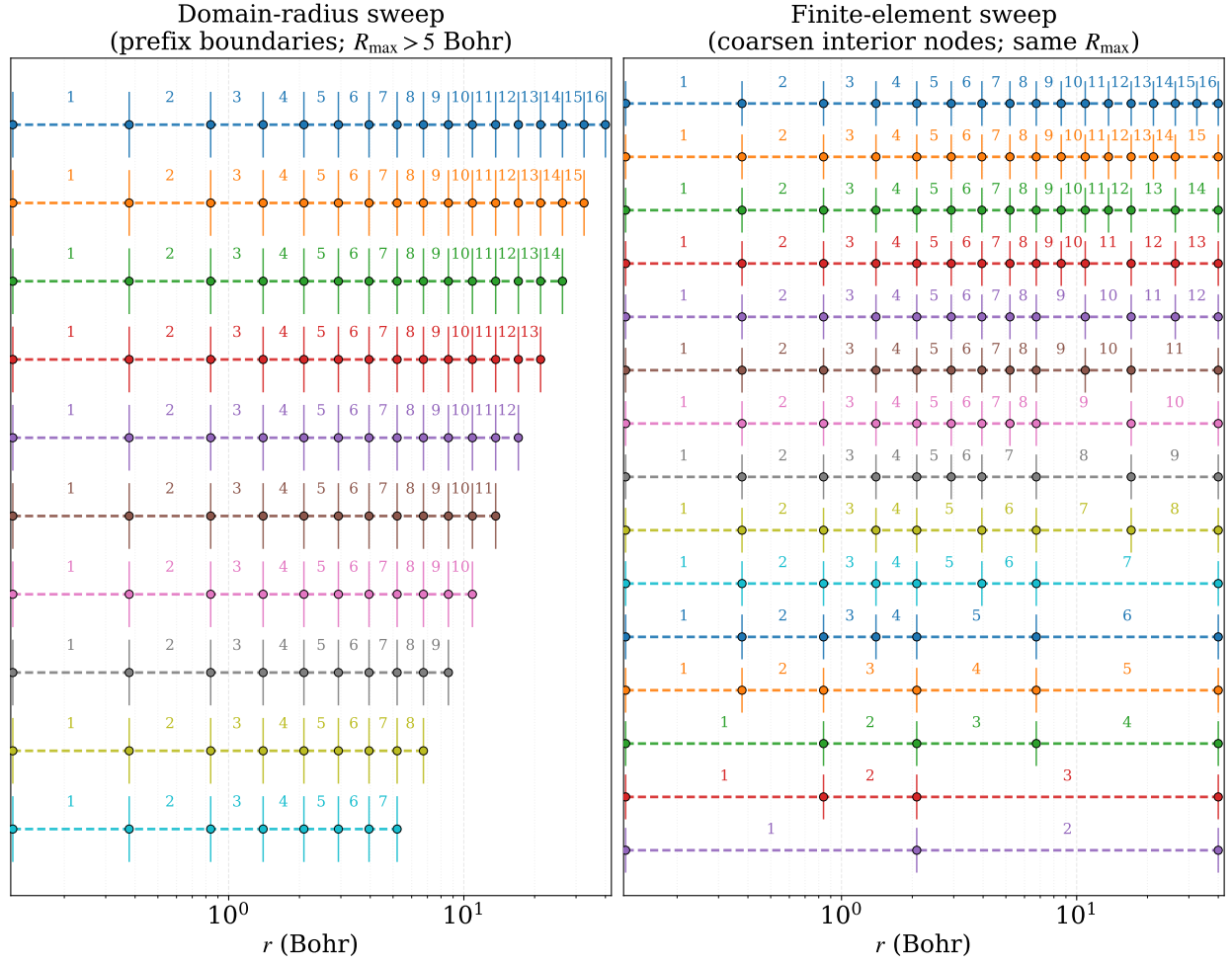


Figure 2: Nested radial mesh subsets used in pseudopotential convergence sweeps: each coarser level retains node locations inherited from the reference exponential mesh, so only resolution changes—not the global mesh law.

Figure 1. Together with the mesh construction recorded above, these results provide additional numerical evidence for the pseudopotential convergence picture presented in the main paper.

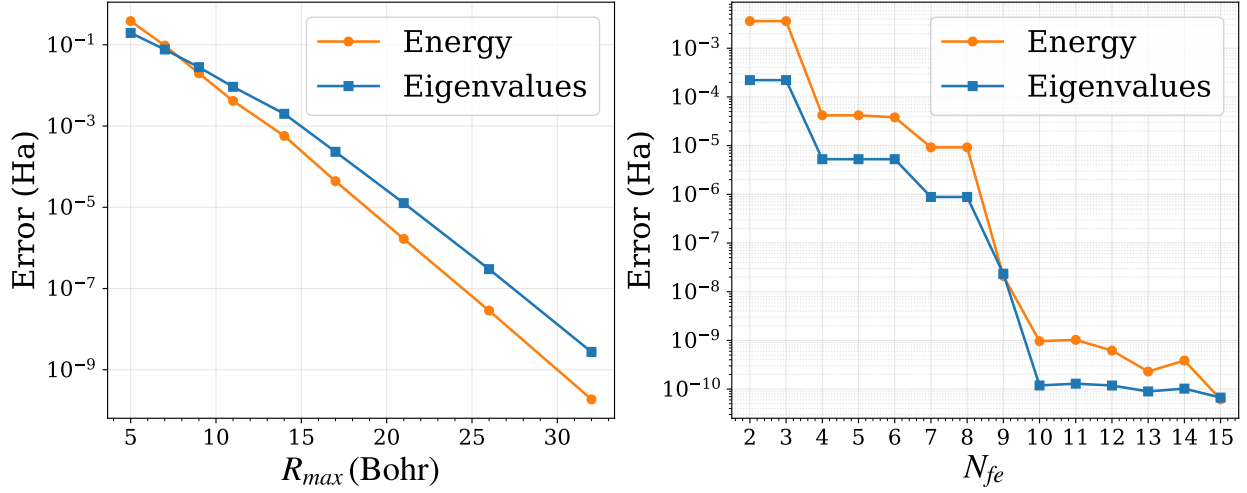


Figure 3: Same as Figure 1, for the LDA-SVWN exchange-correlation functional.

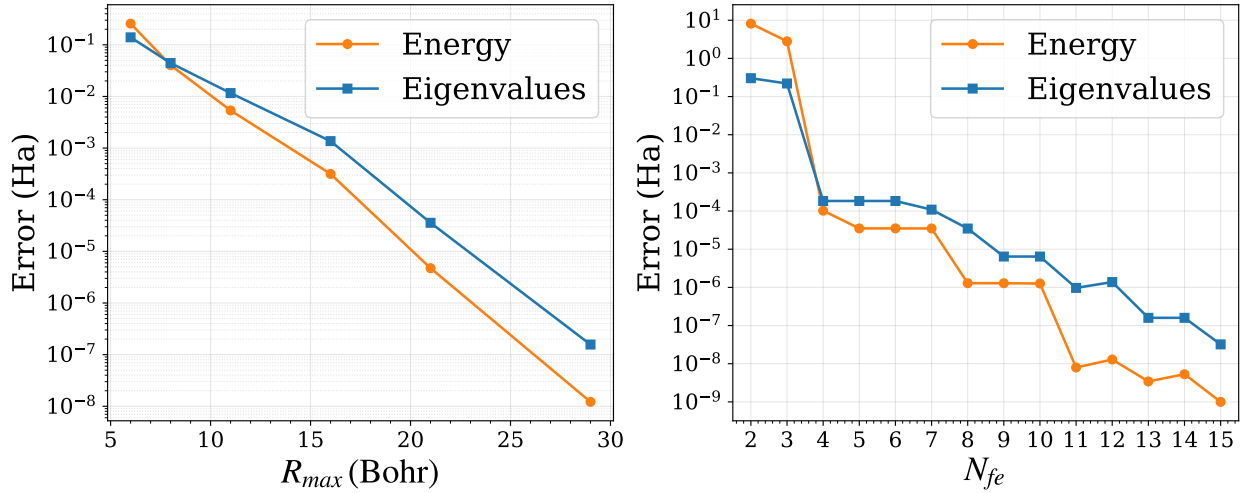


Figure 4: Same as Figure 1, for the rSCAN meta-GGA exchange-correlation functional.

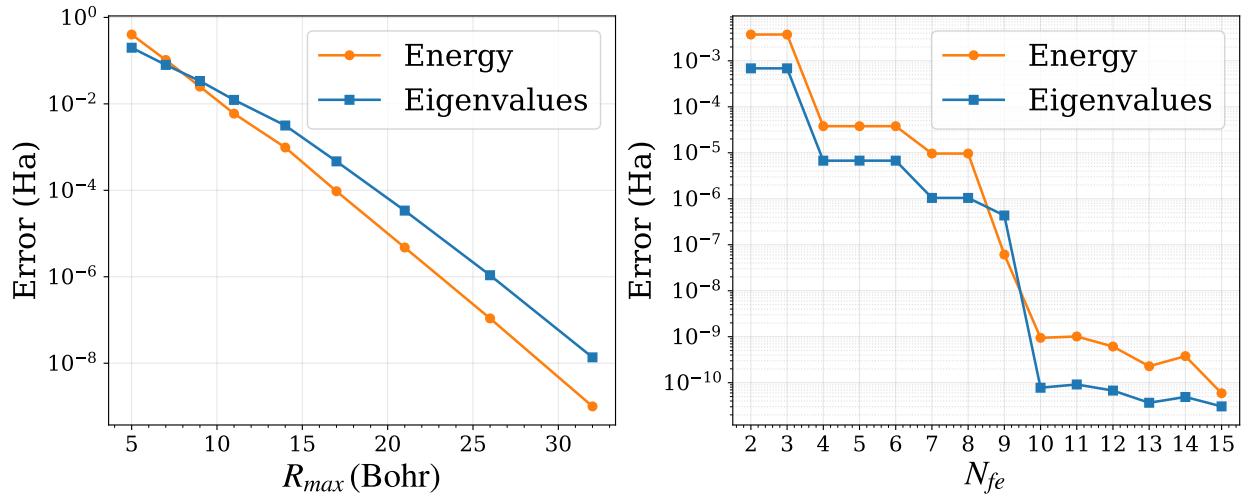


Figure 5: Same as Figure 1, for the PBE0 hybrid exchange-correlation functional.